

Supplementary Material: Adaptive Action Updates in Offline Reinforcement Learning for Integrated Energy-Efficient Building Control

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A Offline Dataset Construction Details

This appendix provides the complete exploration, action-repetition, transition-storage, and preprocessing details for the offline dataset described in Section 4.4 of the main paper.

A.1 Subsystem-Specific Expert Exploration

Let the nominal expert action be

$$a_t^E = \mu_E(o_t),$$

where μ_E denotes the expert controller. The exploratory proposal used in episode e is

$$\tilde{a}_t^E = \mathcal{N}_{\eta_e} \left(a_t^E \right), \quad (1)$$

where $\mathcal{N}_{\eta_e}$ is a subsystem-specific perturbation operator and η_e identifies the exploration profile. Four profiles were used: clean, low, medium, and high.

A.1.1 Glazing Exploration. On steps at which the expert updates the glazing, exploration is triggered with probability ϵ_g . It then modifies one window (probability 0.8) or two windows. For each selected window, the explored state is drawn uniformly from the states within one tier of the currently applied state, excluding the expert’s choice. During glare (work-plane illuminance of at least 1,000 lx), states lighter than the current state and lighter than the glare state are excluded, so exploration never lightens the glazing in glare.

A.1.2 Lighting Exploration. During occupied hours, and on steps at which the expert updates the lighting, Gaussian noise is applied to the dimming fraction of lighting zone z with probability p_t^{noise} :

$$\tilde{\ell}_{t,z}^E = \ell_z^{\text{max}} \text{clip} \left(\ell_{t,z}^E / \ell_z^{\text{max}} + \epsilon_{t,z}^\ell, 0, 1 \right), \quad \epsilon_{t,z}^\ell \sim \mathcal{N} \left(0, \sigma_\ell^2 \right), \quad (2)$$

where ℓ_z^{max} is the installed lighting power of zone z . In perimeter zones, the perturbed power is further capped so that daylight plus artificial illuminance does not exceed 1,000 lx. Defining the noise on the dimming fraction accounts for differences in installed lighting power among zones.



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Table S1: Glazing and lighting action-value exploration parameters.

Profile	ϵ_g	p_t^{noise}	σ_ℓ
Clean	0.00	0.00	0.00
Low	0.05	0.15	0.03
Medium	0.10	0.30	0.05
High	0.20	0.40	0.08

The glazing and lighting exploration parameters are reported in Table S1.

A.1.3 Thermal Exploration. Thermal exploration depends on whether the zone air temperature lies inside or outside the comfort band (21–24 °C, 22–25 °C for episode 013). Inside the band, each zone selects among three behaviours, which are then held for six steps: idle setpoints (16/28 °C), a random inactive band (a heating setpoint below and a cooling setpoint above the current temperature, each with a margin of 0.25 K), and an active target drawn uniformly from 22–24 °C for both setpoints. Outside the band, the corrective target of 22 °C is perturbed with probability $p_{\text{out}}^{\text{noise}}$ by zero-mean Gaussian noise and clipped to 22–24 °C. Leaving the band immediately ends a held behaviour.

The corresponding parameters are reported in Table S2, where p_{rand} is the probability of the random inactive band and p_{target} that of the active target. For each profile, $p_{\text{idle}} + p_{\text{rand}} + p_{\text{target}} = 1$.

Here, $p_{\text{out}}^{\text{noise}}$ is the probability of perturbing a corrective target when the temperature lies outside the comfort band, and σ_T is the standard deviation of the temperature perturbation.

A.2 Probabilistic Action Repetition

Action-value exploration determines the candidate action $\tilde{a}_t^E$, whereas temporal exploration independently determines whether each candidate component is applied.

Let $q(i)$ identify the subsystem associated with action dimension i :

$$q(i) \in \{\text{glazing, lighting, thermal}\}.$$

A repetition decision is sampled independently for each glazing and lighting dimension and jointly for the two setpoints of a zone, the latter only while the zone temperature is inside the comfort band:

$$\begin{aligned} \kappa_{t,i} &\sim \text{Bernoulli} \left(p_{q(i),e}^{\text{rep}} \right), \\ b_{t,i} &= 1 - \kappa_{t,i}, \end{aligned} \quad (3)$$

where $\kappa_{t,i} = 1$ denotes retention and $b_{t,i} = 1$ denotes an update. The executed action is consequently constructed according to Section 4.4 of the main paper. This $b_{t,i}$ describes only the forced repetition; the

Table S2: Thermal action-value exploration parameters.

Profile	p_{idle}	p_{rand}	p_{target}	$p_{\text{out}}^{\text{noise}}$	σ_T [°C]
Clean	1.00	0.00	0.00	0.00	0.00
Low	0.85	0.10	0.05	0.10	0.25
Medium	0.70	0.20	0.10	0.20	0.50
High	0.55	0.30	0.15	0.30	0.70

update mask stored in the dataset and used for training is recomputed from the executed actions (Section 4.4 of the main paper), so it is also zero when the expert itself reproduces its previous value.

The repetition probabilities were set per episode and range from 0 to 0.8. Table S5 lists the resulting forced-repetition rates, measured in each episode. Thermal repetition is only possible inside the comfort band. The action-value exploration profile and the repetition configuration were assigned independently.

A.3 Episode Composition and Reproducibility

The pooled dataset contains 14 complete annual EnergyPlus episodes. Each episode contains 52,559 transitions at an approximately ten-minute control resolution. The episode assignments are

$$\begin{aligned}
 \mathcal{E}_{\text{clean}} &= \{000\}, \\
 \mathcal{E}_{\text{low}} &= \{001, 002\}, \\
 \mathcal{E}_{\text{medium}} &= \{003, \dots, 010, 012\}, \\
 \mathcal{E}_{\text{high}} &= \{011, 013\}.
 \end{aligned} \tag{4}$$

Each episode used its own deterministic exploration seed. The seeds are listed in the repository.

A.4 Transition Representation

For each timestep, the stored transition is

$$\mathcal{T}_t = (o_t, a_{t-1}^{\text{exec}}, a_t^{\text{exec}}, b_t, c_t, r_t, o_{t+1}, d_t, e_t), \tag{5}$$

where d_t denotes episode termination and e_t identifies the source episode and exploration profile.

The stored proposal-conditioning vector is

$$c_t = (1 - b_t) \odot a_{t-1}^{\text{exec}} + b_t \odot \xi, \quad \xi = -2 \cdot 1. \tag{6}$$

All normalised physical actions lie in $[-1, 1]$; therefore, $\xi = -2 \cdot 1$ is an out-of-range marker. For a retained dimension, $c_{t,i}$ contains the previous executed value. For an updated dimension, it contains the marker value. Although c_t is stored in the dataset, it is deterministic given a_{t-1}^{exec} and b_t .

Annual episodes remain contiguous in the pooled dataset. The terminal indicator is set only for the final transition of each episode, producing 14 terminal transitions. No artificial daily or weekly episode boundaries are introduced, and recurrent training sequences are not permitted to cross annual episode boundaries.

A.5 Shared Preprocessing Statistics

Observation normalisation follows Section 4.1 of the main paper. The means and standard deviations of the 23 physical observation variables are computed from all 735,826 transitions in the pooled offline training dataset. The six bounded temporal and occupancy

Table S3: State-independent selectors at k times the learned per-subsystem update rates or at the behaviour controller's rates (RBC rate). Same checkpoint and proposals, means over selector seeds. Bold: best per column.

Selector	Budget	kWh/d	Viol. %	Kzh/d	Vis. %	Upd. %
Learned (sampled)	–	344.1	7.4	1.23	48.4	13.7
Periodic	1× learned	301.6	22.0	5.79	27.7	13.6
	2× learned	328.5	18.9	5.07	48.6	27.2
	4× learned	363.1	14.6	3.18	58.7	54.4
	RBC rate	339.8	17.8	4.68	29.1	26.9
Random	1× learned	296.9	23.9	6.48	20.9	13.6
	2× learned	314.4	19.8	5.11	33.3	27.2
	4× learned	357.8	14.7	3.03	48.9	54.3
	RBC rate	313.9	19.6	5.09	19.3	26.9
Time of day	1× learned	310.3	22.4	5.90	44.4	13.6
Always update	–	359.8	14.7	2.73	48.8	100.0

variables, corresponding to dimensions 24 through 29, are passed through without standardisation.

Reward-component scales are also computed from the pooled training dataset. For raw reward component j , the shared scale is

$$s_j = \frac{1}{N} \sum_{t=1}^N |r_{t,j}^{\text{raw}}|. \tag{7}$$

A single set of observation and reward statistics is used for all expert and exploration profiles. These statistics are fixed during offline optimisation and subsequent policy evaluation.

B Reward Weights

After division by its mean absolute value over the offline dataset, each reward component is multiplied by the following weight: thermal comfort 3; perimeter visual under-illumination 1.5; HVAC electric demand 1; core under-illumination 1; lighting energy 0.4; controllable perimeter over-illumination 0.4; core over-illumination 0.4; artificial-light excess 0.25; daylight over-illumination 0.2; solar-gain penalty while cooling 0.2; solar-gain bonus while heating 0.1; update penalty 0.1. Before normalisation, the update penalty charges 0.05 per glazing, 0.01 per lighting and 0.002 per setpoint update. Components are clipped to ± 10 (± 50 for thermal comfort), summed and passed through symlog. The visual-comfort variant (Section 6.3 of the main paper) uses the same weights except controllable and daylight perimeter over-illumination, both raised to 1.

Table S4: Observation history: $K=36$ (ours) and $K=1$ (one training run). Mean $\pm$ half-range over selector seeds. Δ : $K=1$ minus $K=36$, paired by day, 95% moving-block bootstrap interval. Bold: better value.

History	kWh/d	Viol. %	Kzh/d	Vis. %	Upd. %
$K=36$ (ours) ($n=5$)	344.1 $\pm$ 3.2	7.43 $\pm$ 1.06	1.23 $\pm$ 0.29	48.4 $\pm$ 1.1	13.7
$K=1$ ($n=3$)	344.6 $\pm$ 1.1	4.89 $\pm$ 0.04	0.54 $\pm$ 0.01	46.1 $\pm$ 0.6	15.7
Δ	+0.5 [-0.5, +4.6]	-2.54 [-3.23, -1.74]	-0.68 [-1.00, -0.28]		

Table S5: Exploration profile and measured forced-repetition rates (glazing / lighting / thermal) of the 14 data episodes.

Episode	Profile	Forced repetition (%)
000	clean	0 / 0 / 0
001, 002	low	5 / 5 / 8
003, 004	medium	5 / 5 / 8
005	medium	2 / 2 / 3
006	medium	0 / 0 / 0
007	medium	20 / 20 / 8
008	medium	40 / 40 / 26
009	medium	60 / 60 / 45
010	medium	80 / 80 / 54
011	high	10 / 10 / 4
012	medium	10 / 10 / 5
013	high	10 / 10 / 3

C Update-Rate Sweep

Table S3 lists the values plotted in Figure 2 of the main paper (Section 6.2).

D Observation History

Table S4 compares the reported agent ($K=36$) with the same method trained with a one-step history ($K=1$; one training run, three selector seeds). Electricity is unchanged, and thermal violations and their severity are lower without the history. Because this is a single training run and the violation rate of our agent varies by about 1.7 pp (SD) across its five training runs, we read it as evidence that the history is not needed in this building rather than as an improvement.

E Baseline Training Length

Stopping the three original baselines of Table 4 of the main paper at 50 instead of 100 epochs does not help: their first training runs then have 27.4% (full-action IQL), 29.6% (full-action BC) and 29.5% (act/repeat BC) occupied thermal violations, against 25.0%, 29.2% and 27.6% after 100 epochs (means over five training runs). The gap to our agent is therefore not caused by training the baselines too long.